

DESIGN PROJECT

TUTORIAL SHEET

BTECH and BEng Level 3 | Academic Year 2025–2026

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Exercise 1

The President of IUGET has tasked a team of engineers with launching a new BTech programme in Cyber security at the Douala campus. This initiative has never been undertaken before at the institution.

- (a) Define the term project and identify FIVE key characteristics of a project. Illustrate each characteristic using evidence from the scenario above.
- (b) Distinguish clearly between a project and routine operations, giving THREE points of difference. Use a table in your answer.
- (c) Explain the concept of the Triple Constraint (Iron Triangle) and discuss how it applies to the scenario described above.

Exercise 2

The Government of Cameroon, in partnership with the African Development Bank (AfDB), is planning to extend the national electricity grid to 50 previously unelectrified rural communities in the Adamawa Region.

- (a) Describe any FOUR standard phases of the project cycle, clearly stating the purpose, key activities, and outputs of each phase.
- (b) Compare the AfDB/World Bank project cycle with the EU Project Cycle Management (PCM) model. State THREE similarities and TWO differences.
- (c) Explain why project cycle models are important for development organisations such as the AfDB.

Exercise 3

The Urban Community of Douala (CUD) has identified that the city's main drainage infrastructure is severely inadequate, causing seasonal flooding across 14 neighbourhoods, destroying property, disrupting economic activity, and leading to cholera outbreaks. The proposed project involves the

construction of 85 km of primary and secondary drainage channels, 12 retention basins, and 6 motorised pumping stations, as well as the rehabilitation of 38 km of flood-affected urban roads. Total estimated cost: XAF 68 billion. Duration: 5 years. Financing: World Bank Urban Resilience Programme and CUD. The project requires acquisition of 4.2 km² of land.

- (a)** Apply the Problem Tree technique to the Douala flooding situation described above. Clearly identify: (i) the Core Problem; (ii) at least FOUR root causes; and (iii) at least FOUR negative effects on the city and its population.
- (b)** Describe the SIX disciplines of project management that the CUD project manager must apply to successfully deliver this drainage infrastructure project. For each discipline, state: (i) a clear definition; and (ii) one specific application to the Douala drainage project.
- (c)** Construct a Risk Register for the Douala drainage project. Identify FIVE risks specific to this urban infrastructure project. For each risk, state: the risk type, its probability (High/Medium/Low), its potential impact, and the recommended response strategy. Use template below

No.	Risk Description	Type	Probability	Potential Impact	Response Strategy
1.					
2.					

Exercise 4

An NGO called EnergyAccess Cameroon wishes to design and implement a solar mini-grid project to provide electricity to 3,000 households and 20 health centres in the Mbam-et-Inoubou Division of Centre Region. The project will be submitted for funding to the World Bank over a period of 3 years with a proposed budget of XAF 2.5 billion.

- (a)** List and briefly describe EIGHT sections that must appear in the project proposal for EnergyAccess Cameroon. For each section, state its purpose and one key element.
- (b)** Construct a Logical Framework (Logframe) matrix for the EnergyAccess Cameroon solar mini-grid project. Your Logframe must include: Goal, Purpose, at least TWO Outputs, and at least FOUR Activities — each with at least one indicator, means of verification, and key assumptions.
- (c)** Using the six OECD-DAC evaluation criteria, explain how the EnergyAccess Cameroon project would be evaluated after completion. For each criterion, state what specific question an evaluator would seek to answer.